

A First Look at Starlink’s Geofeeds: On Discrepancies between Feeds and Routing

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Abstract

Geofeeds (RFC 8805) play a decisive role in the determination of the IP-Geolocation, while carrying a high uncertainty about accuracy and error rate. LEO satellite networks exacerbate the problem, since user location and Point of Presence assignment differ. Starlink publishes two separate feeds, trying to solve the uncertainty problem, creating two different answers for a geolocation with the consequence of implausible pairings and customers not knowing which feed to choose for their intended use. We aim to deliver a method for the validation of the geofeed information against active routing measurements. We probe the two geofeeds with a combination of anycast and traceroute measurements to find inconsistencies, monitor change events and try to find a way to predict changes. The central contributions of this paper are: an analysis of the past development and feed dynamics of the two geofeeds; our initial measurement results indicating first routing-based error and disagreement rate between the two geofeeds.

1 Introduction

Today IP geolocation information is relevant to various use cases, such as information security or content delivery. To deliver accurate locations in 2020, RFC 8805 was published, defining a format for operators to self-report authoritative geolocation data [12]. RFC 9632 later defined how such geofeed data is discovered [4]. Since then, consumers have ingested public geofeeds as an important data source and rely on them for operational decisions, e.g., mapping traffic to the closest Point of Presence (PoP) of a CDN [14]. However, geofeed coverage remains sparse, and published data is often erroneous [14]. Geolocation providers such as IPinfo find self-reported geolocation information to be “often the root cause of bad IP geolocation data” [10]. Thus, they validate this information using measurements from a globally distributed probe measurement network [11]. The standard and its consumers disagree on whether self-reported information is always accurate. This challenge already exists in static networks, and is even more relevant in dynamic ones such as LEO satellite networks, whose geofeeds have not yet been surveyed in depth.

Geofeeds rely on a terrestrial assumption: The user’s location is physically close to their egress PoP. LEO satellite networks break this assumption [15]. Starlink publishes two feeds: a user-location feed following RFC 8805 [18], and a PoP feed mapping prefixes to the terrestrial egress sites in its network [19]. The two feeds show discrepancies [18, 19], as the snapshot of 2026-08-25 00:40 shows. 399 prefixes carry an RFC location with no PoP assignment,

and several prefixes show implausible pairings of RFC location and assigned PoP. For example, Sudan’s 32 prefixes all read *Khartoum* while egressing via six PoPs on two continents.

Accurate geofeed information and a defined error rate are crucial for several use cases: APNIC Labs uses geolocation data to attribute ISP market share across countries [1]. After the feed led APNIC’s algorithm to assign 60% of Yemen’s Internet user population to Starlink, APNIC overrode the feed and marked 20 economies unclassified [9]. In content delivery, Bose et al. show that moving PoPs closer to users cuts median page-fetch time by 60% [3]. A recent LEO measurement study of Starlink usage patterns by Kravtsov et al. explicitly names the geolocation lag as a limitation [13]. All three use cases need to know how accurate the feeds are, and no such error rate exists: no prior work validates either feed against actual routing. Geofeed studies to date are cross-sectional and terrestrial [6, 14]. The rare Starlink geofeed work is anecdotal, describing the problem without quantifying it [15].

We present a measurement study of both feeds, tracked continuously and extended with archival snapshots back to 2021. We validate the feed data against real routing measurements. We identify at least two responsive IPs per prefix and probe them continuously from the 32 globally distributed nodes of an anycast testbed [8, 17]. Additionally, we use IPinfo’s ProbeNet to run traceroutes from probes worldwide to our target IPs [11]. Our main contributions in this work are:

- **Feeds vs. routing:** First measured disagreement rates between the two feeds and observed routing.
- **Feed error characterization:** Characterization of feed errors and their silent correction.
- **Early PoP shift detection:** Potential routing-based detection of PoP shifts before the feed publishes them.

2 Methodology

Feed collection. We continuously tracked and documented changes in both, the `feed.csv` (RFC 8805, user location) [18] and `pops.csv` (prefix to egress PoP) [19] since June 2026 at an hourly rate, relying on the HTTP ETag response header [7] to identify unchanged versions, to build a continuous dataset for further investigation. Polling hourly gives us a timestamp for all snapshots, independent of the `about.json` feed timestamp, which was previously updated on a schedule rather than on change and does not distinguish between the two feeds [15]. Therefore, we can bracket later-discussed asymmetric and symmetric changes, e.g., the PoP and location change of prefix 143.105.58/24, to ≈ 23 h. To enable long-term analysis of

both feeds, our dataset is supplemented by snapshots from the community geofeed archive [20], dating back to January 2024 for the feed.csv and to the first publication of the pops.csv in October 2025, and snapshots from the Internet Archive for the feed.csv before 2024, dating back to April 2021.

Change taxonomy. We classify the changes of the two different feeds into three distinct categories:

- **Added:** A prefix was added to the feed that did not exist in the previous snapshot in any way.
- **Deleted:** A prefix which was included in the previous snapshot is not included in the feed anymore in any way.
- **Reassigned:** A prefix location or PoP changed in comparison to the previous snapshot, this also includes prefixes for which a split or aggregation occurred simultaneously.
 - **Split:** A missing prefix from the previous snapshot, being replaced by two or more prefixes covering the same address space (e.g., one /24 prefix $\Rightarrow$ two /25 prefixes).
 - **Aggregated:** An added prefix covering a previously-split address space (e.g., two /25 prefixes $\Rightarrow$ one /24 prefix).

Target selection. To compare the feed data and dynamics with real-world routing, we select probing targets by consecutively scanning the IPv4 prefixes in feed.csv with ZMap, using TCP-SYN packets [5], due to their higher response rate and usage in another part of the setup. We choose feed.csv due to it being a superset of the pops.csv. We try to find ≥ 2 responsive IP addresses per prefix as probing targets. We initially scan the whole feed.csv once, the responsive IP addresses are used as a hitlist and scanned for 14 days, consecutively reducing the unresponsive IP addresses from it. Our hitlist covers $\geq 40\%$ of the feed.csv and $\geq 45\%$ of the pops.csv.

Routing measurement – breadth. We employ two separate setups for the routing measurement. In our first setup, we use the anycast testbed TANGLED [2] from which we probe our full hitlist with TCP-SYN packets from 32 globally distributed nodes using a measurement system developed for LACeS [8]. The target replies via the BGP-nearest node, except when more distant links are used due to, e.g., economic policies, revealing changes and anomalies in the geofeed’s location assignment of the probed prefixes. The semidiurnal full hitlist coverage is fast enough to resolve the rate at which the feeds themselves change, 0.60-0.69 changes per day. With combined feed snapshots and the Anycast measurement, we aim to find routing anomalies, prove the efficacy of this method for predicting changes before they are documented in the geofeed, and calculate the corresponding lag time. IP addresses that go unresponsive during our measurement are therefore invisible in this measurement setup.

Routing measurement – depth. Complementing the breadth measurement above, we zoom in on individual change events using traceroutes from IPinfo’s ProbeNet probes [11]. IPinfo’s ProbeNet includes 1,395 probes in 173 countries. We use a subset of 76 probes, with a minimum of one probe per the 56 Starlink PoPs and 20 randomly sampled to ensure geographic diversity. Every 12 hours, we conduct traceroutes to a fixed number of IPs, sampled from the responsive hitlist gathered in the target selection, due to measurement capacity and scheduling, covering all Starlink PoPs. Each run yields full paths, hop counts, and reachability traces, granting

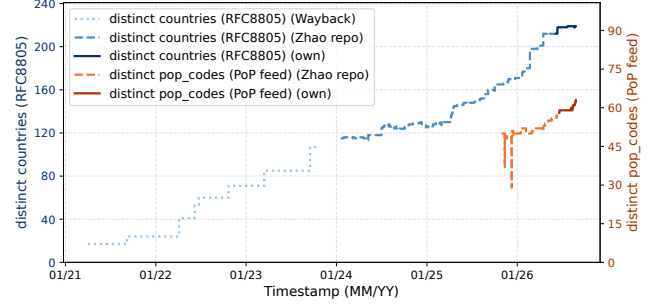


Figure 1: Distinct countries in the RFC 8805 feed against distinct PoP codes in the PoP feed over time.

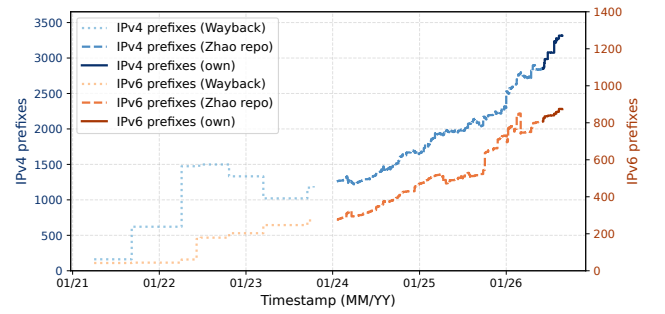


Figure 2: IPv4 and IPv6 prefix counts in the RFC 8805 feed over time, by data source.

an in-depth look at possible location changes and trace failures exposing the transition itself.

Ethics and responsible measurement. To ensure responsible measurements, we limit ourselves to ethical probing rates (max. 100 packages per second), ensure opt-out is possible by providing a webpage on the IP address we use to scan with ZMap, and save no user-identifying data as recommended for ZMap usage [5].

3 Evaluation

Change type	RFC avg/day	PoP avg/day
Added	5.14	18.91
Deleted	1.93	15.67
Reassigned	0.95	0.86
Aggregations	0.20	0.34
Splits	0.14	0.00

Table 1: Average prefix changes per calendar day, RFC 8805/PoP feed observed over 968/329 days.

Feed composition and growth. Looking at the snapshot from 2026-08-25 00:40 of the two feeds, the feed.csv covers 4,192 prefixes across 197 countries and 16 US states, while the pops.csv covers 3,793 prefixes across 63 distinct PoP codes. Next to the asymmetry in granularity, 399 prefixes ($\approx 9.5\%$) appear in the feed.csv

while not being included in the `pops.csv`, no such cases exists in the reverse; past snapshots confirm this is a structural gap. Since the more frequent logging of changes in the feed in January 2024 and the launch of the `pops.csv` in October 2025, both feeds show a clear growth in prefixes and locations/PoP codes as visualized in Figure 1. Figure 2 visualizes the growth of IPv4 and IPv6 prefixes in the `feed.csv`: the number of IPv4 prefixes more than doubled ($\approx 1,250$ in January 2024 to $\approx 3,300$ in August 2026), and IPv6 prefixes show a similar development. The Wayback archive and our own collected data are visible as separate series; our own measurement and the archive are easily combined due to the overlapping timeframe, while the Wayback data leaves an unavoidable gap between itself and the archive data.

Feed dynamics. While the RFC 8805 feed `feed.csv` exists longer on the internet, the PoP feed `pops.csv` changes more often with a cumulative churn of 0.69 against 0.60 changes per day on the `feed.csv` in the time frame since 2026-01-01. A per-day change volume by class, as described in the Methodology, is listed in Table 1 presenting a majority of additions and occasional deletions including also rare but large reassignment events.

We also found a clustering of changes made in both feeds around 17:00-02:00 UTC, US Pacific business hours on Monday to Friday with a minimum on Sunday, indicating the feeds are maintained by hand. This corroborates the qualitative claim of previous work with timing data and explains the error classes below [15].

Feeds vs. routing. To validate the impact of changes in the routing table, we use the traceroutes facilitated by ProbeNet probes to further analyze a specific reassignment event. The IPv4 prefix `143.105.58.0/24` was reassigned in both feeds between 2026-07-27 21:59 and 2026-07-28 20:59, changing the PoP from `tmpeazx1` (Phoenix, US) to `c1gyca1` (Calgary, CA) and the RFC 8805 location from US/US-AZ/Phoenix to CA/CA-SK/Regina. The corresponding ProbeNet traceroutes show an independent, but agreeing shift in the ingress IP address, as well as a change in the last ASN passed before ingress from the probe in Calgary at the newly assigned PoP, skipping the ASN 6461 and routing directly into the Calgary IX before entering Starlink's ASN. Furthermore, the traces in the time frame of 2026-07-28 09:00-21:00 failed, indicating unresponsiveness during the period the shift was holding, i.e., a durable reassignment. The traces establish that the technique aligns with real-world changes and represent a concrete instance of the cross-border activation, a user located in a country being served by infrastructure in another, that previous work [13] flagged as a limitation.

Another phenomenon is the implausible pairing of RFC location and corresponding PoP assignment. Figure 3 visualizes which PoP covers the majority of prefixes located in a country or US state according to the two feeds in the timeframe of our measurements from 2026-07-15 until 2026-08-25. Despite an active PoP in Stockholm, SE, $\approx 33\%$ of prefixes with the RFC location in the `feed.csv` in Stockholm, SE, were assigned to the PoP in Frankfurt, DE for months. Another case shows two prefixes with an even larger discrepancy, with the RFC location being in Dallas, US, while the `pops.csv` locates the prefixes at the PoP in Almaty, Kazakhstan. A similar case pairs the Frankfurt, DE PoP with the majority of prefixes with an RFC location in Jerusalem, IL, as visible in Figure 3. The first case was resolved by reassigning the corresponding prefixes to the mentioned PoP in Stockholm, SE; still, the assignment

existed for months in both feeds. The second case was handled by a complete deletion of the relevant prefixes from both the `feed.csv` and `pops.csv`. After being first added in early July 2026, the anomaly still persisted for ≈ 2 months and also was clearly separable from the other prefixes assigned to the PoP in Almaty, since the four genuine prefixes with their RFC location in Kazakhstan map correctly to the Almaty PoP (`almykaz1`). The third case persists for our whole measurement period. Contrary to the prevailing assumption that traffic routed to an IPv4 address is answered through the PoP location the prefix is assigned to [16], our measurements reveal that the traffic sometimes seems to end up at the RFC location instead. In the case of the Anycast setup having a vantage point at the RFC location and the assigned PoP location, $\approx 5\%$ of responsive IPs in our hitlist routed their answer to the VP at the assigned RFC location, rather than the PoP location.

Lead and lag. Our Anycast setup holds the potential for being more than additional input in our historic analysis of geofeed changes. We were able to identify a PoP reassignment of a prefix located at `frntdeu1` (Frankfurt, DE) to `lndngbr1` (London, GB) ≥ 48 hours before the change was published in the `pops.csv`. Furthermore, for the two prefixes that are only in the `feed.csv` and have responsive IPs, we predict the PoP feed entries to be at `lagax1` (Atlanta, US) for the prefix `192.31.243.0/24` located in Ashburn, US-VA and `sttlwax1` (Seattle, US) for the prefix `129.224.216.0/24` located in Dutch Harbor, US-AK. We treat this as a direction for future work, not a validated result.

4 Discussion

Which location should a consumer use? The user location and PoP assignment that Starlink maps in `feed.csv` and `pops.csv` are both correct answers to different questions, and consumers pick inconsistently: Ookla follows the `feed.csv` while Cloudflare and Google serve at the PoP. The initial results of our measurement setup show that in the clear majority ($\approx 95\%$) of the cases where an anycast VP was at the RFC and PoP location, the `pops.csv` seems to be agreeing with the anycast routing.

Consequences for geolocation databases. The problem of inaccuracy in geofeeds was illustrated, for example, by APNIC Labs' response, discarding the feed for ≈ 20 economies [9]. Our per-prefix measurement covers 11 of the 20 economies included in the discard. For subset A (e.g., Zimbabwe, Madagascar), routing patterns are consistent with the feed's claim, suggesting Huston's per-country override discarded valid data. For subset B (Yemen, Sudan, South Sudan), all probed prefixes routed to European VPs 5000+ km from the claimed location, corroborating Huston's concerns. Additionally, feeds carry no timestamps and no TTL. Combining this with our measured churn rates, a consumer polling the `feed.csv` and the `pops.csv` weekly misses ≈ 25 and holds ≈ 7 outdated RFC locations, as well as missing ≈ 124 and holding ≈ 7 outdated PoP assignments. This is a direct, quantifiable cost of the missing feed metadata, independent of whether individual assignments are correct at publication time.

Recommendations. Our initial measurement results show that Starlink's feeds are a lagging indicator of its own network. Our active measurement and routing-based validation can be used as a leading one and a standing check on any self-published feed,

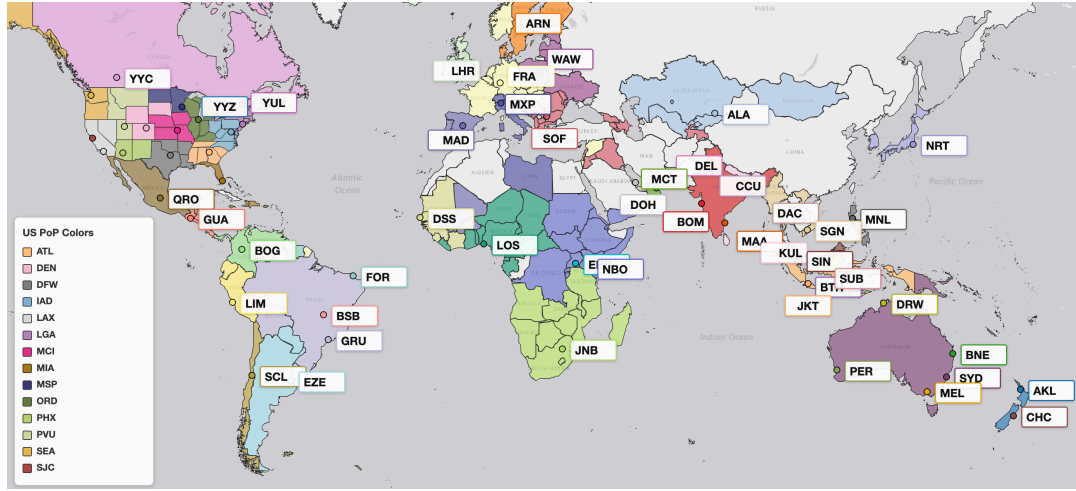


Figure 3: Geographic distribution of dominant Starlink PoP to RFC coverage

not only Starlink's. To address these complications, we support the recommendations of an embedded timestamp and TTL, as proposed in [15], plus guidance on granularity. The timestamp merely gives users an indication of how old the feed is at the time it is fetched, whereas a TTL could lead to customers adapting the fetch interval to the change rate. For Starlink's feeds explicitly, we recommend publishing the pops.csv in a documented, standardized form, or folding the PoP assignment into the RFC 8805 record.

Limitations. Although our measurement setups show promising first results, the method needs refinement and time to produce more sophisticated data. Our findings are Starlink-specific for now, but we are confident our methodology generalizes to other geofeed publishers; the observed error rates and anomalies obviously do not. Our anycast validation only involves responsive targets. The silent majority of the feeds is out of scope for this setup. Even for responsive targets, we resolve the reply-catching VP rather than the exact user location. The significance of our measurement results is also limited by the density of anycast VPs. The historical feed data before our own polling comes from third-party and Wayback archives with unknown intermediate coverage.

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